

Lecture 3: Wireless Physical Layer: Modulation and Coding Techniques

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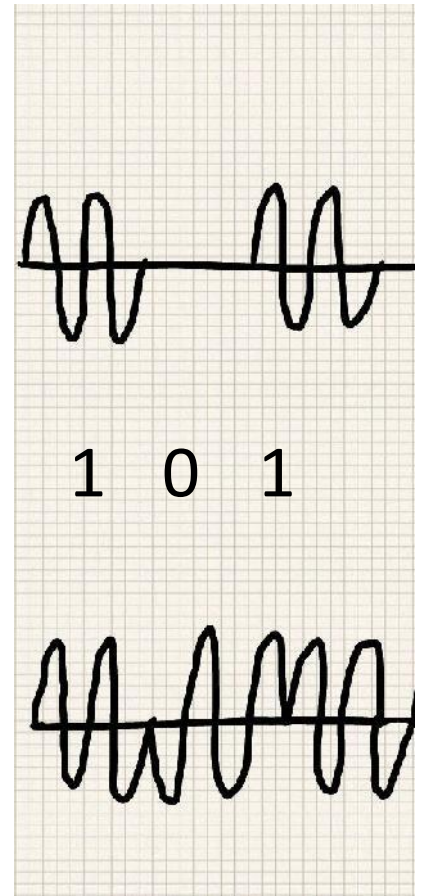
Ack: Mythili Vutukuru

Modulation

- We saw a simple example of amplitude modulation in the last lecture
- Modulation – how to transmit a stream of bits using a carrier wave of a particular frequency and a certain bandwidth
- Carrier wave $s = A \cos (2 \pi f t + \varphi)$
- Can modulate one or more of the following to transmit bits
 - Amplitude A
 - Frequency f
 - Phase φ
- We will cover only a high level overview of modulation techniques in this lecture

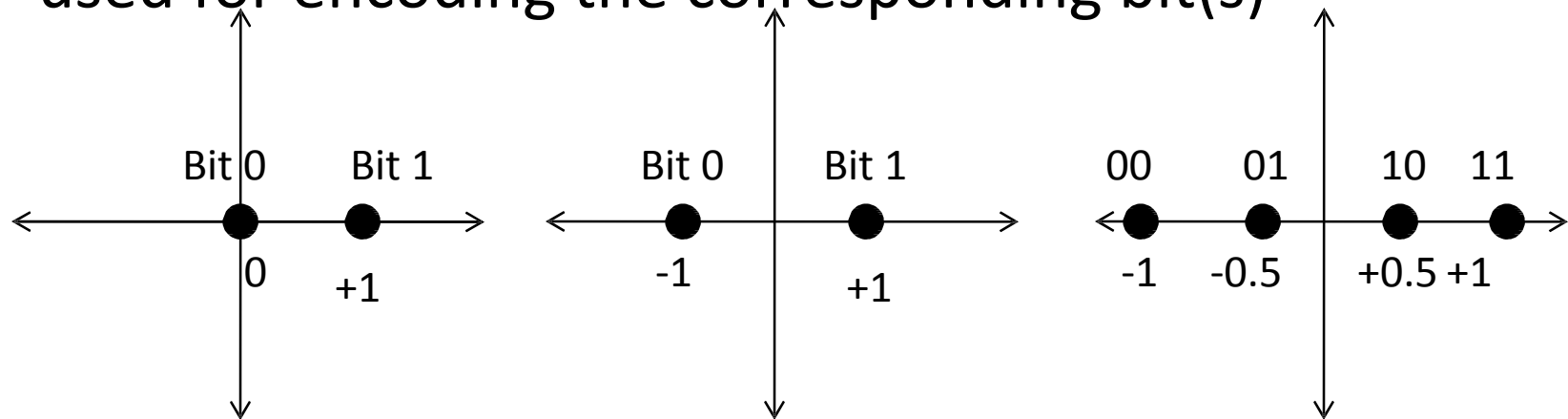
Amplitude Shift Keying (ASK)

- Use amplitude of 0 for bit “0” and 1 for bit “1”. This is called 2-ASK.
- Note that the actual amplitude depends on the transmit power, we will use 1 to denote the maximum A
- Or, use -1 for bit “0” and +1 for bit “1”. Amplitude -1 means that the wave is “inverted”
- We can encode multiple bits. For example, 4 different amplitude values to convey 2 bits: 00, 01, 10, 11. (4-ASK)



Constellation diagrams

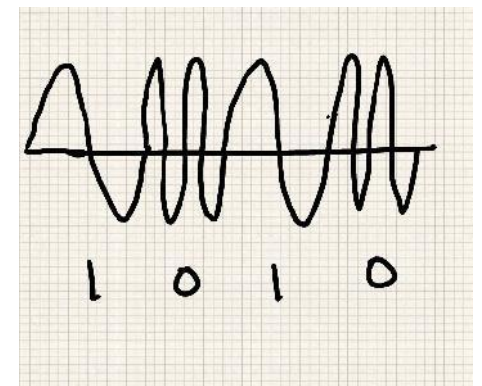
- Easy way to represent modulation schemes instead of drawing waveforms
- Value on x-axis determines the amplitude of wave used for encoding the corresponding bit(s)



- The above constellation diagrams show two different 2-ASK schemes and one 4-ASK scheme

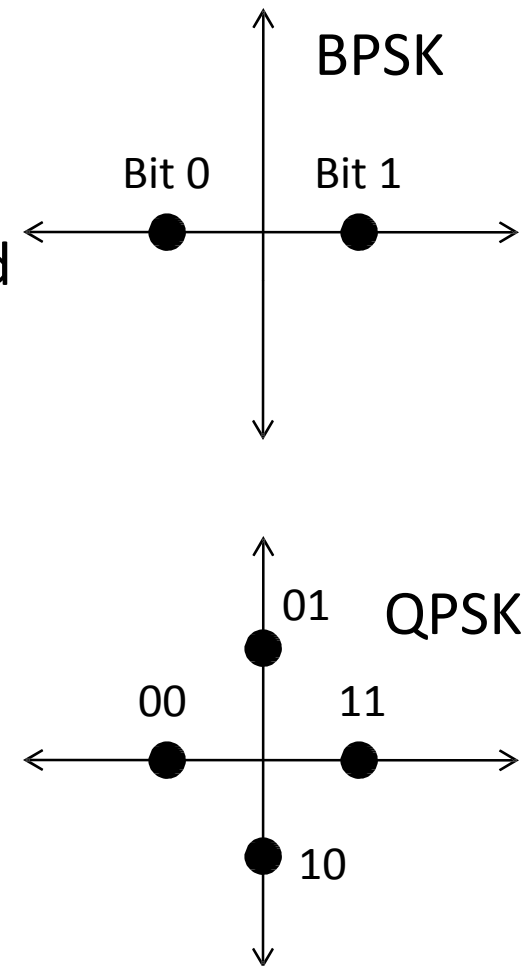
Frequency shift keying (FSK)

- Use two different frequencies to transmit bit 0 and bit 1 (binary FSK)
- Can also send multiple bits
- Not very widely used, as it consumes more bandwidth than other techniques



Phase Shift Keying (PSK)

- Use different phases of the wave to send bits
- Binary PSK (BPSK) – phase 0 and phase π to send bits 0 and 1. Looks like 2-ASK with amplitudes -1 and +1
- QPSK (quaternary PSK): use 4 phases to send 2 bits
- For PSK constellation diagrams, the radial angle indicates phase, distance from origin indicates amplitude (always 1 for PSK)

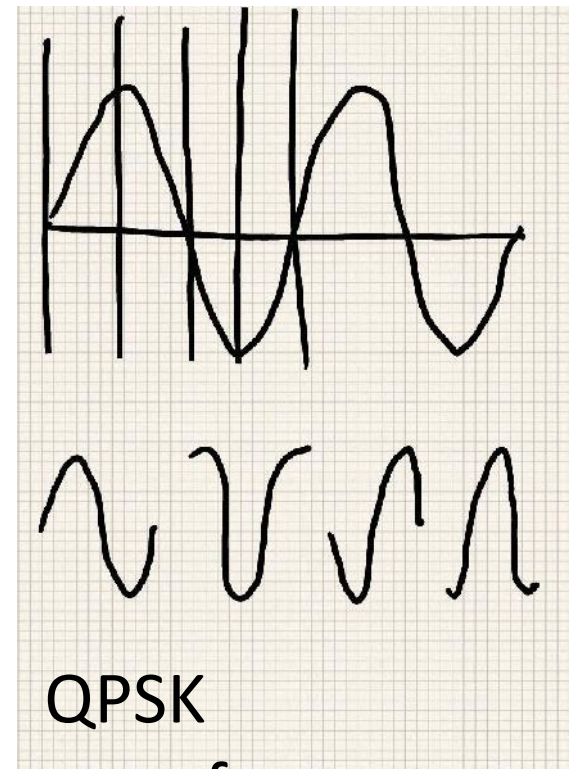


PSK (2)

- Example of QPSK wave forms. 4 different phases to send bits 00, 01, 10, 11
- A symbol is a set of two bits that map to a wave form
- Transmitter stitches together waveforms for each symbol
- 8-PSK is also possible, but inefficient. Not

widely used

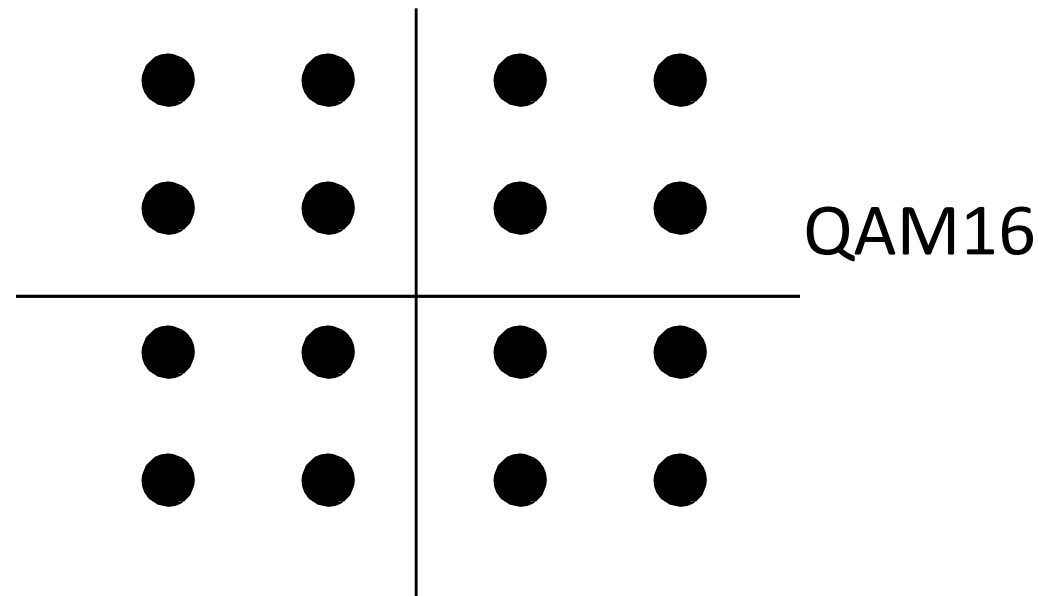
- PSK needs “phase lock” between transmitter and receiver to estimate phase at receiver
- Another idea: **differential QPSK** (DQPSK) – just take the difference in phase between previous and current symbol to convey information. No phase lock.



QPSK
waveforms

Quadrature Amplitude Modulation (QAM)

- Use both amplitude and phase to send information
- QAM16, QAM64 etc – widely used for high speed in mobile systems
- Denser QAM constellations require higher SNR to decode correctly

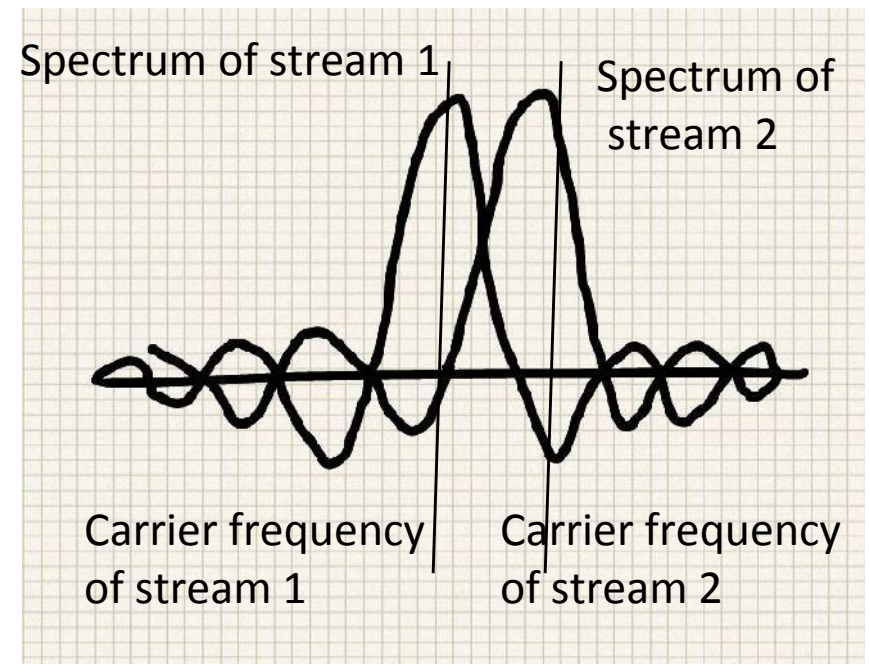
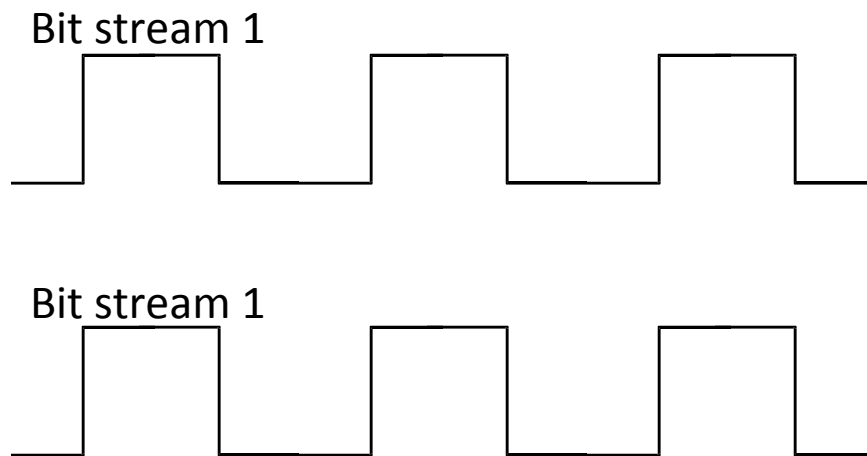


Single Carrier Modulation vs. Multi-carrier Modulation

- So far, ASK, PSK, QAM etc are all examples of single carrier modulation schemes – one stream of bits modulating one carrier signal
- How fast can we send – depends on bandwidth, sampling rate of hardware etc (last lecture)
- Note that **most modulation techniques require knowing exact amplitude and phase of carrier**, so we must compensate effect of **channel “h” (Equalization)**
- If each **symbol duration is longer than multipath delay spread** of channel, all copies of a symbol arrive in the same symbol duration. **Easier to estimate the channel “h”** and compensate for it at receiver
- If **symbol duration < delay spread**, copies of previous symbol interfere with current symbol – inter-symbol interference (**ISI**)
- When large delay spread causes ISI, we need complicated equalization techniques to cancel out effects of all previous symbols (“multi-tap equalization”) at receiver
- **Single carrier** systems – tradeoff between how fast you can send and how complex your receiver can be. **Not very suitable** for high rates in compact mobile devices
- Solution – **multi-carrier modulation**

Multi-carrier modulation

- Split bit stream into multiple parallel streams. Modulate each stream with different carriers within the allocated band. Send each stream slowly, so that no ISI happens. Recover each stream separately at receiver.
- Will the parallel streams not interfere? It is possible to choose slightly different carrier frequencies for each stream, such that the carriers don't interfere (i.e., when each carrier is at peak, all other carriers are zero).



Orthogonal Frequency Division Multiplexing (OFDM)

- This technique is called OFDM. The standard modulation technique in almost all high speed systems today.
- Split channel into multiple subcarriers (e.g., 64 subcarriers in 802.11/WiFi)
- Send a parallel stream of data over each subcarrier “slowly” (relative to delay spread)
- At receiver, only simple equalization (“single-tap equalization) required
- Can recover multiple streams of data simultaneously at receiver

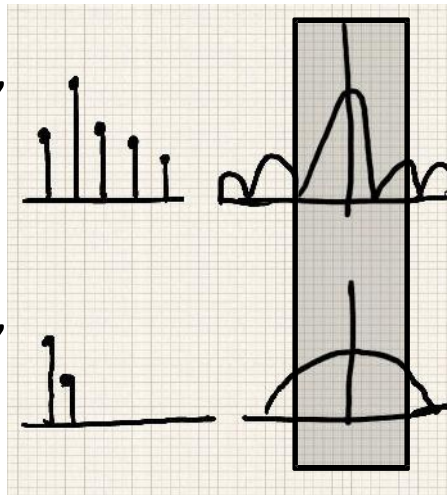
OFDM (2)

- Receiver is simple, but transmitter is very complex? Generate multiple carriers and modulate, then add up?
- There is an easier way to do it using fourier transform.
- Recall, DFT takes a time domain signal and converts it into weights (amplitudes and phases) of each of the individual frequencies. Inverse DFT reverses this process.
- At OFDM transmitter, for each symbol, take the bits from each parallel stream, map to amplitude and phase (PSK or QAM modulations). Now, take these amplitude and phase of the N subcarriers, perform inverse fourier transform to get time domain signal, then modulate this signal alone at center frequency.
- **Fast Fourier Transform (FFT)** algorithm is efficient implementation of DFT. FFT and invert FFT (iFFT) hardware implementations make OFDM very easy to implement.

Frequency domain view of OFDM

- Why does OFDM work?
- Channel impulse response “ h ” $\rightarrow$ its DFT is called channel frequency response “ H ” (captures attenuation of each frequency in the band)
- With large delay spread, frequency response of the channel varies within the band. At receiver, we need to estimate the complex shape of the “ H ” curve to “invert” the channel.
 - And vice versa. For low delay spread, channel frequency response stays the same over the entire band. See figure below.
- With OFDM, each stream has its own narrow band, where H stays almost the same. So need to estimate only one value of “ H ” for each sub-band. Easier to do.

Channel impulse response “ h ”
with large delay spread



Channel impulse response “ h ”
with small delay spread

Channel frequency response “ H ”
varies a lot in the frequency band

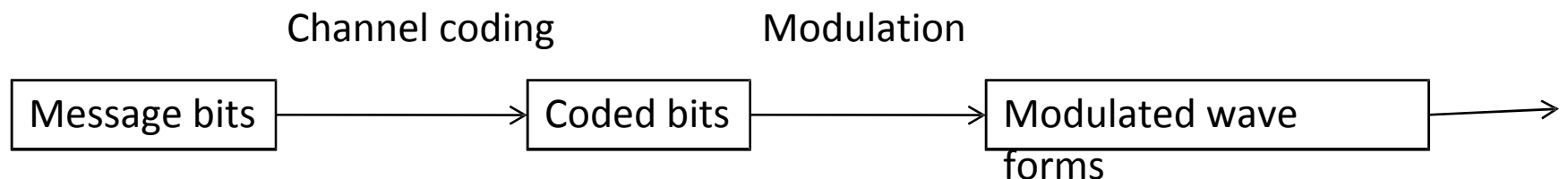
Channel frequency response “ H ”
almost constant in the band

Channel Coding

- Modulated waveforms disrupted by signal propagation through wireless channel – leads to errors when recovering bits from signal
- Channel coding – add redundancy in the transmitted bits, so that you can detect and correct errors
- **Error detection** – received coded bits indicate that bit errors have occurred
- **Error correction** – received coded bits indicate that bit errors have occurred, and help to identify and correct the location of errors

Modulation and Channel Coding

- The modulation and coding schemes decide the **bit rate** of transmissions over the air.
- Wise choice of modulation and coding schemes helps reach Shannon capacity
- Message bits are converted into coded bits before modulation at transmitter
- At receiver, the received waveform is demodulated into bits, then the channel code is decoded to recover the original message bits



Simple example – repetition coding

- A very simple channel coding scheme is repetition coding – repeat each bit multiple times, so that you can recover the message even if some bits are in error
- For bit “0” – send “000”
- For bit “1” – send “111”
- “000” and “111” are called valid codewords
- Received codeword should ideally be 000 or 111. In reality, due to errors, it can be any 3 bit string, say 001
 - If we assume only 1-bit errors, we can detect and correct the error and say that the transmitted code word was 000
 - If we assume 1 or 2 bit errors, then we know there was an error, but we can't be sure what the transmitted codeword was (only error detection, no correction)
- That is, this simple code can detect up to 2 bit errors, or correct single bit errors

Bit rate of a transmission

- Each OFDM symbol has 64 samples
 - 64 subcarriers, after iFFT, result in 64 time samples
- On a 20MHz channel, we can send 20M samples per second
- At 20M samples/sec, each symbol takes 3.2 microseconds
- A 0.8 microsecond guard time added to each symbol
- Therefore, each OFDM symbol takes 4 microsec.
- Channel delay spread of the order 100ns, so very little ISI
- If single carrier modulation were used, note that symbols have had to be 64 times shorter

Bit rate of a transmission (2)

- Bit rate depends on modulation and coding scheme used
- For example, QAM16 and rate $\frac{1}{2}$ code
- Each OFDM symbol has 48 data subcarriers -> 48×4 coded bits -> 48×2 data bits
- 96 bits in one symbol in 4 microsec -> bit rate is $96/4 = 24$ Mbps
- Similarly, we can get all bit rates in 802.11g (6, 9, 12, 18, 24, 36, 48, 54 Mbps) by various combinations of modulation (BPSK, QPSK, QAM16, QAM64) and coding rates ($\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$)

Bit rate of a transmission (3)

- Higher bit rates require higher SNR to work properly
- For example, if SNR is 10dB, 6Mbps and 9Mbps rates may have 0% loss, 12 Mbps rate has 10% loss, 18 Mbps has 50% loss, and 24Mbps and higher have 100% loss. What is the best bit rate to use?
- Clearly, 12 Mbps has higher throughput = $\text{bit_rate} * \text{packet_delivery_ratio}$
- We will study how bit rates are picked in WiFi in detail later in the course
- For now, understand where the numbers in the rates come from.

Modulation and Coding => Data Rate

- 802.11g

Modulation	Coding	DataRate (Mbps)
BPSK	1/2	6
BPSK	3/4	9
QPSK	1/2	12
QPSK	3/4	18
16QAM	1/2	24
16QAM	3/4	36
64QAM	2/3	48
64QAM	3/4	54

MIMO – new idea

- Recent physical layer designs (e.g., newer WiFi standards like 802.11n, and cellular systems like LTE) use a new concept called multiple-input-multiple-output (MIMO) to improve physical layer rates further
- In MIMO, you have multiple antennas at transmitter and receiver
- You can use multiple antennas in two ways
 - Send and receiver multiple copies of the same signal, to increase chances of successful reception (transmit / receive diversity mode)
 - Send multiple parallel streams of data (spatial multiplexing mode)
- Notation: 2X2 MIMO means 2 transmit antennas and 2 receive antennas

MIMO (2)

- Receive diversity – if you have multiple antennas at receiver, you can receive multiple copies of the signal and combine them in an optimal way. This way, you can get lower errors.
- Transmit diversity – send multiple copies of the signal in a clever way so that they combine constructively at receiver
- Spatial multiplexing – send multiple parallel streams of information between sender and receiver. If you have N transmit antennas and N receive antennas, you can theoretically send N parallel streams of data between sender and receiver
- For example, consider the 54 Mbps rate (in a single antenna system). If used in spatial multiplexing mode in 2X2 MIMO configuration, we can get a rate of 108 Mbps

Putting it all together: transmission

- We will describe how a WiFi / 802.11 physical layer works. Let's take the example of 802.11g
- 20 MHz bandwidth in the 2.4Ghz band is used
- The physical layer transmits one frame (obtained from higher MAC layer) at a time
- CRC is added to the message to enable error detection
- Bits in a packet converted to coded bits
- Bits split into 48 parallel streams for OFDM (OFDM uses 64 subcarriers, but only 48 carry data)
- In each subcarrier stream, bits are grouped into symbols based on the modulation scheme used (BPSK has 1 bit per symbol, QPSK has 2, QAM16 has 4, QAM64 has 6)

Putting it all together: transmission (2)

- Each group of bits is modulated using the corresponding subcarrier
 - Group of bits mapped to amplitude/phase values of the subcarrier wave (using constellation diagrams)
 - The vector of 64 such amplitude/phase values are passed through iFFT to get a 64-sample time domain frequency by modulating with a carrier at 2.4 GHz
- A physical layer header describing the modulation, coding, frame length etc is added to the start of the packet
- A special preamble symbol is added to the start of the frame. The preamble is a known set of bits modulated by BPSK, so all WiFi nodes know what the signal looks like

Putting it all together: reception

- Receiver is always searching the radio waves at the given frequency for the special known preamble
- When the preamble is found, the receiver detects start of the frame and starts decoding the samples
- The wireless channel h is estimated from the preamble and compensated on all subsequent samples of the packet
- The receiver takes each OFDM symbol, splits it into subcarriers (by using FFT), uses the amplitude and phase of the subcarrier to demodulate the transmitted bits
- The demodulated bits are passed through a channel decoder to recover the original message bits from the decoded bits
- Finally, CRC is checked to see if all bits received correctly
- Correct frames are passed up to higher layers